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STM32 – Encoder

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Incremental encoders

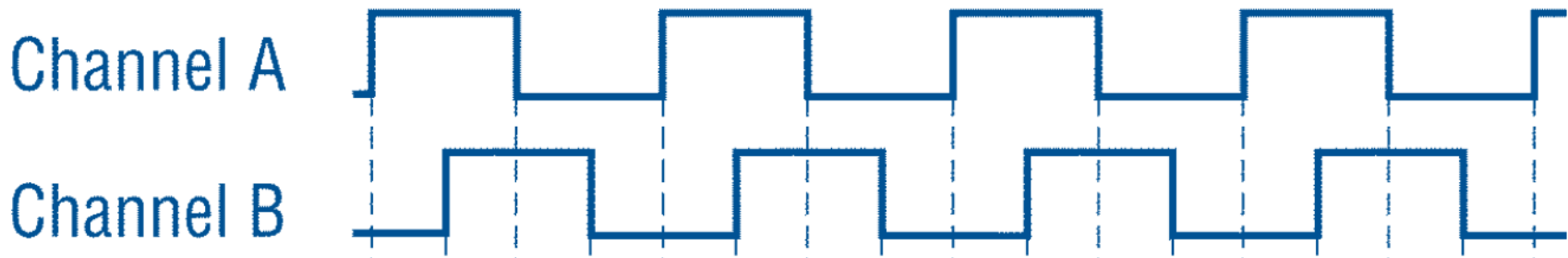
Provide a specific number of pulses per revolution (PPR) in rotary motion, or per inch or millimeter in linear motion.

- **single channel output** → doesn't provide direction of movement
- **quadrature output** → provides direction sensing
(two channels 90° out of phase)

To determine position, its pulses must be accumulated by a counter.

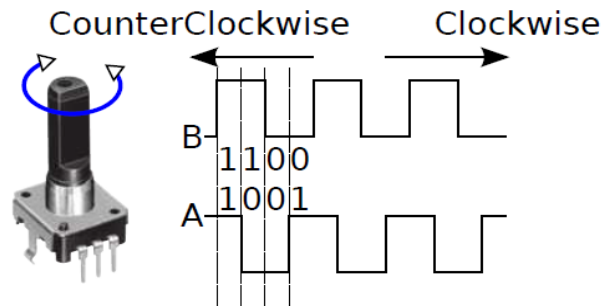
When starting up, the equipment must be driven to a reference or home position to initialize the position counters.

Some incremental encoders also produce another signal, the “marker,” produced once per revolution.





Encoder modalities

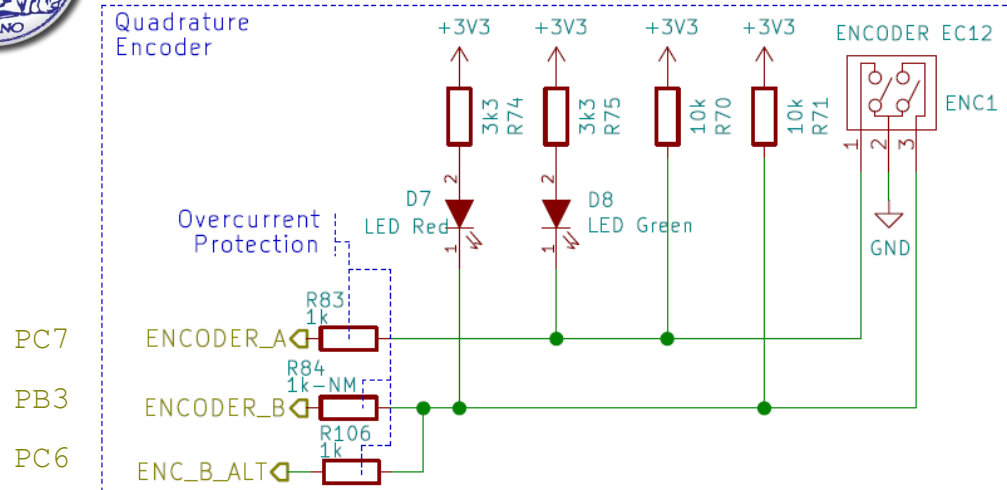


Active edge	Level on opposite signal (TI1FP1 for TI2, TI2FP2 for TI1)	TI1FP1 signal		TI2FP2 signal	
		Rising	Falling	Rising	Falling
Counting on TI1 only	High	Down	Up	No Count	No Count
	Low	Up	Down	No Count	No Count
Counting on TI2 only	High	No Count	No Count	Up	Down
	Low	No Count	No Count	Down	Up
Counting on TI1 and TI2	High	Down	Up	Up	Down
	Low	Up	Down	Down	Up

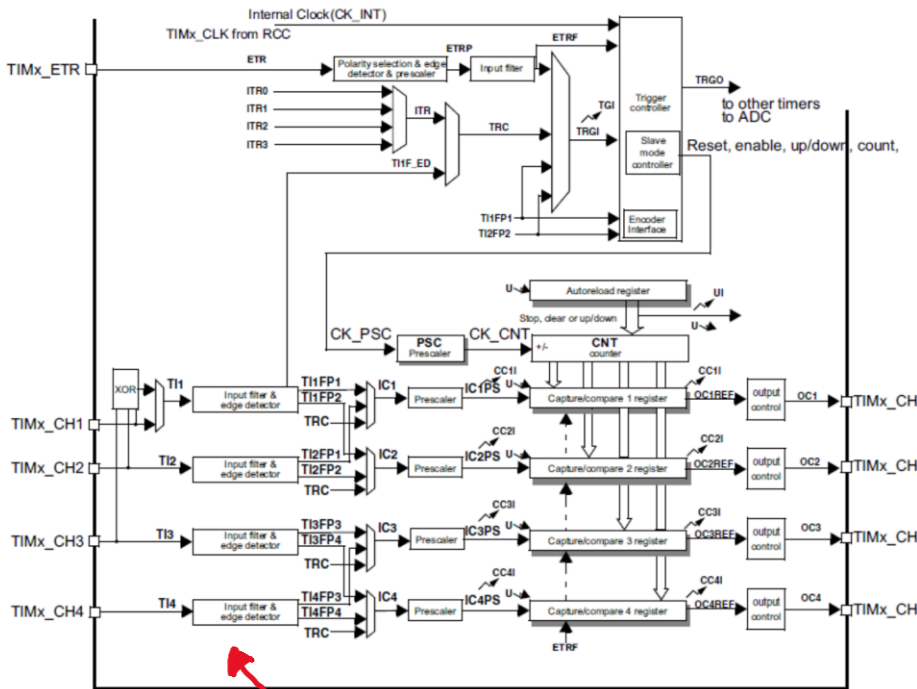
- **Encoder mode TI1:** the timer's counter is updated on the A signal edges, depending on the B signal level. Every mechanical step of the encoder results in a ± 2 increase of the counter, so a whole revolution of the shaft results in a ± 24 increase of the counter.
- **Encoder mode TI2:** the timer's counter is updated on the B signal edges, depending on the A signal level. Every mechanical step of the encoder results in a ± 2 increase of the counter, so a whole revolution of the shaft results in a ± 24 increase of the counter.
- **Encoder mode TI1 and TI2:** the timer's counter is updated on both A and B signal edges, depending on the other channel's stable level. Every mechanical step of the encoder results in a ± 4 increase of the timer's counter, so a whole revolution of the shaft results in a ± 48 increase of the counter.



PMDB16: encode

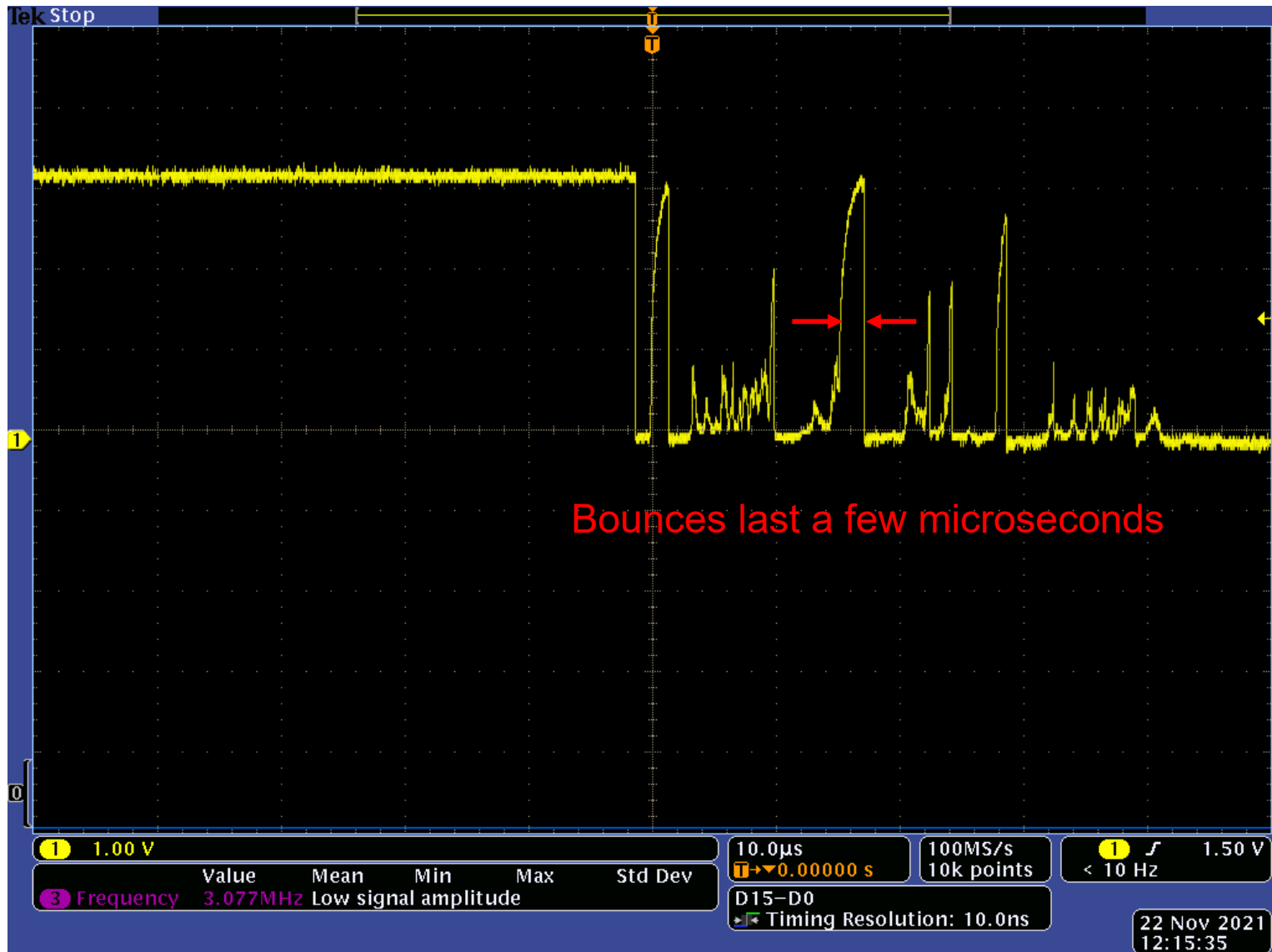


- The encoder is connected to pins PC6 / PC7 of the STM32
- STM32 Timer peripherals feature a dedicated encoder mode: pins PC6 and PC7 can only be controlled by TIM3
- No hardware debouncing: we will use **digital filtering**





PMDB16: encoder debouncing





STM32 Timer input digital filter

IC1F: Input capture 1 filter

This bit-field defines the frequency used to sample TI1 input and the length of the digital filter applied to TI1. The digital filter is made of an event counter in which N consecutive events are needed to validate a transition on the output:

0000: No filter, sampling is done at f_{DTS}

0001: $f_{SAMPLING} = f_{CK_INT}$, N=2

0010: $f_{SAMPLING} = f_{CK_INT}$, N=4

0011: $f_{SAMPLING} = f_{CK_INT}$, N=8

0100: $f_{SAMPLING} = f_{DTS}/2$, N=6

0101: $f_{SAMPLING} = f_{DTS}/2$, N=8

0110: $f_{SAMPLING} = f_{DTS}/4$, N=6

0111: $f_{SAMPLING} = f_{DTS}/4$, N=8

1000: $f_{SAMPLING} = f_{DTS}/8$, N=6

1001: $f_{SAMPLING} = f_{DTS}/8$, N=8

1010: $f_{SAMPLING} = f_{DTS}/16$, N=5

1011: $f_{SAMPLING} = f_{DTS}/16$, N=6

1100: $f_{SAMPLING} = f_{DTS}/16$, N=8

1101: $f_{SAMPLING} = f_{DTS}/32$, N=5

1110: $f_{SAMPLING} = f_{DTS}/32$, N=6

1111: $f_{SAMPLING} = f_{DTS}/32$, N=8

Minimum pulse duration

@ $f_{CK_INT} = 84 \text{ MHz}$, CKD = 1 (CKD = 4)

Unfiltered: 11.9 ns (47.7 ns)

381 ns (1.52 μs)

3.05 μs (12.2 μs)

$$f_{DTS} = f_{CK_INT} / \text{CKD (Internal clock division)}$$



STM32CubeIDE: setup

TIM3 Mode and Configuration

Mode

Channel3 Disable

Channel4 Disable

Combined Channels Encoder Mode

☐ Use ETR as Clearing Source

☐ XOR activation

Configuration

Reset Configuration

✓ NVIC Settings ✓ DMA Settings ✓ GPIO Settings

✓ Parameter Settings ✓ User Constants

Configure the below parameters :

Search (Ctrl+F)

Counter Mode Up

Counter Period (AutoReload Register .. 65535

Internal Clock Division (CKD) Division by 4

auto-reload preload Disable

▼ Trigger Output (TRGO) Parameters

Master/Slave Mode (MSM bit) Disable (Trigger input effect not delayed)

Trigger Event Selection Reset (UG bit from TIMx_EGR)

▼ Encoder

Encoder Mode Encoder Mode TI1

____ Parameters for Channel 1 ____

Polarity Falling Edge

IC Selection Direct

Prescaler Division Ratio No division

Input Filter 15

____ Parameters for Channel 2 ____

Polarity Rising Edge

IC Selection Direct

Prescaler Division Ratio No division

Input Filter 15



Project 1 – Encoder readout

Objective

Read the encoder position
and send to the PC the
rotation speed in rpm

Use a timer as a timebase and DMA to transfer the UART data



Aim of the project

The objective of the project is to readout a quadrature encoder using the specific modality of STM32 timers in order to provide the rotation frequency (expressed in rpm/rotations per minute) and direction (“+” for clockwise and “-” for counterclockwise).

The result must be displayed using the remote terminal.

HAL functions:

`HAL_StatusTypeDef HAL_TIM_Encoder_Start(TIM_HandleTypeDef * htim, uint32_t Channel)`

- **htim:** pointer to a TIM_HandleTypeDef structure that contains the configuration information for TIM module.
- **Channel:** TIM Channels to be enabled. This parameter can be one of the following values:
 - TIM_CHANNEL_1: TIM Channel 1 selected
 - TIM_CHANNEL_2: TIM Channel 2 selected
 - TIM_CHANNEL_ALL: TIM Channel 1 and TIM Channel 2 are selected

`__HAL_TIM_GET_COUNTER(TIM_HandleTypeDef * htim)`



Project hints

- Identify the encoder pins, and enable them in TIMx_CHy mode
- Set up the timer to operate in encoder mode, with the correct input filter applied. Start the timer in encoder mode.
- Retrieve the counter value every second and compute the delta from the previous read, then convert it to rpms.
How many counts does a full rotation of the encoder provide?
- Beware of overflow and underflow of the timer. How to solve this issue?
- Compile and debug the code.